JAVA LAB OBSERATION -7

1. Basic Information

Title: Demonstrate 10 Exception Classes in Java with Code Examples

2. Objective / Aim

To write and execute a Java program to demonstrate 10 different exception classes using try-catch blocks and understand how exceptions are handled during program execution.

3. System Requirements

Hardware Requirements:

- Computer or laptop
- Minimum 4 GB RAM
- Keyboard and monitor

Software Requirements:

- Operating System: Windows / Linux
- Java Development Kit (JDK): JDK 8 or later
- IDE: Eclipse, IntelliJ IDEA, NetBeans, or Visual Studio Code
- Text Editor: Notepad or any Java-supported editor

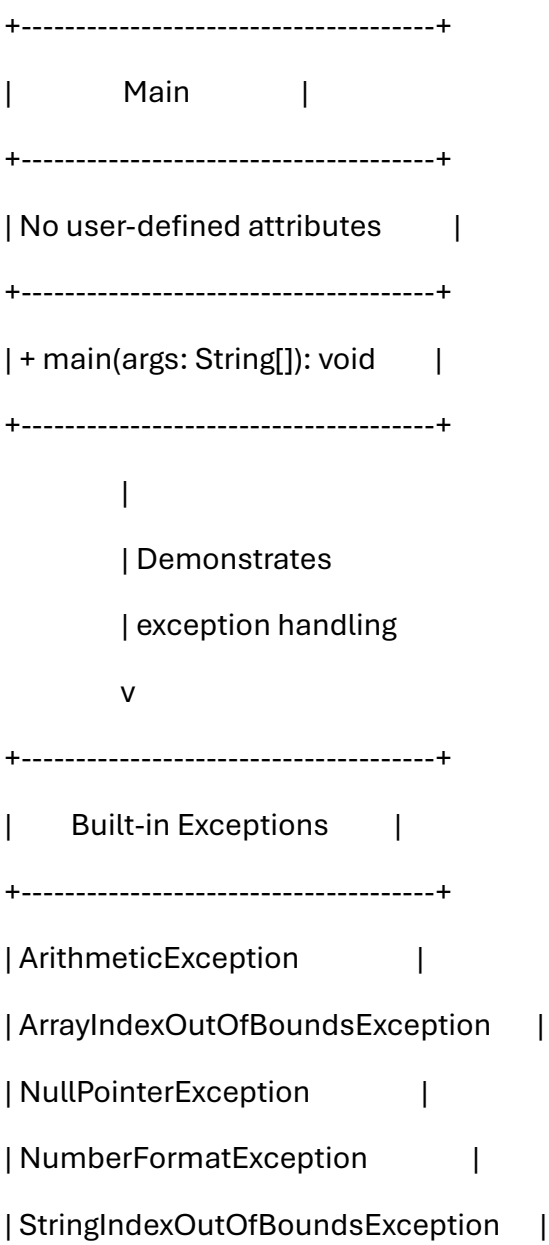
4. Algorithm / Logic Description

1. Start the program.
2. Create a class named Main and define the main() method.
3. Use a try block to perform an operation that generates an ArithmeticException.
4. Use separate try-catch blocks to demonstrate the following 10 exception classes:
    - ArithmeticException
    - ArrayIndexOutOfBoundsException
    - NullPointerException
    - NumberFormatException
    - StringIndexOutOfBoundsException
    - InputMismatchException

- NegativeArraySizeException
- ClassCastException
- FileNotFoundException
- ClassNotFoundException

5. Place the appropriate exception class in each catch block.
6. Print a message when an exception is caught.
7. Continue executing the remaining examples after each exception is handled.
8. Display the output and stop the program.

5. UML Class Diagram



| InputMismatchException |

| NegativeArraySizeException |

| ClassCastException |

| FileNotFoundException |

| ClassNotFoundException |

+-------------------------------------+

Exception Inheritance Hierarchy:

Throwable

|

+-- Exception

|

+-- RuntimeException

| |

| +-- ArithmeticException

| +-- IndexOutOfBoundsException

| | |

| | +-- ArrayIndexOutOfBoundsException

| | +-- StringIndexOutOfBoundsException

| +-- NullPointerException

| +-- IllegalArgumentException

| | |

| | +-- NumberFormatException

| +-- NegativeArraySizeException

| +-- ClassCastException

| +-- NoSuchElementException

| |

| +-- InputMismatchException

|

```
    +-- IOException
    |    |
    |    +-- FileNotFoundException
    |
    +-- ReflectiveOperationException
         |
         +-- ClassNotFoundException

  +-- Error
```

6. Source Code

```
import java.io.*;
import java.util.*;

public class Main {
   public static void main(String[] args) {

      // 1. ArithmeticException
      try {
         int a = 10 / 0;
      } catch (ArithmeticException e) {
         System.out.println("1. ArithmeticException");
      }

      // 2. ArrayIndexOutOfBoundsException
      try {
         int a[] = {1, 2, 3};
         System.out.println(a[5]);
      } catch (ArrayIndexOutOfBoundsException e) {
```

```
    System.out.println("2. ArrayIndexOutOfBoundsException");
}

// 3. NullPointerException
try {
    String s = null;
    System.out.println(s.length());
} catch (NullPointerException e) {
    System.out.println("3. NullPointerException");
}

// 4. NumberFormatException
try {
    int n = Integer.parseInt("abc");
} catch (NumberFormatException e) {
    System.out.println("4. NumberFormatException");
}

// 5. StringIndexOutOfBoundsException
try {
    String s = "Java";
    System.out.println(s.charAt(10));
} catch (StringIndexOutOfBoundsException e) {
    System.out.println("5. StringIndexOutOfBoundsException");
}

// 6. InputMismatchException
try {
```

```
    Scanner sc = new Scanner("hello");
    int n = sc.nextInt();
    sc.close();
} catch (InputMismatchException e) {
    System.out.println("6. InputMismatchException");
}

// 7. NegativeArraySizeException
try {
    int a[] = new int[-5];
} catch (NegativeArraySizeException e) {
    System.out.println("7. NegativeArraySizeException");
}

// 8. ClassCastException
try {
    Object obj = "Hello";
    Integer n = (Integer) obj;
} catch (ClassCastException e) {
    System.out.println("8. ClassCastException");
}

// 9. FileNotFoundException
try {
    FileInputStream f =
        new FileInputStream("missing_file_123.txt");
} catch (FileNotFoundException e) {
    System.out.println("9. FileNotFoundException");
```

```
        }

        // 10. ClassNotFoundException
        try {
            Class.forName("NoSuchClass12345");
        } catch (ClassNotFoundException e) {
            System.out.println("10. ClassNotFoundException");
        }
    }
}
```

7. Compilation and Execution Commands

Step 1: Save the Program

Save the source code with the filename:

Main.java

Step 2: Compile the Program

Open Command Prompt or Terminal and enter:

javac Main.java

Step 3: Execute the Program

Enter the following command:

java Main

Note: Ensure that Java is installed and configured correctly on your computer.

8. Output Screen

1. ArithmeticException
2. ArrayIndexOutOfBoundsException
3. NullPointerException
4. NumberFormatException
5. StringIndexOutOfBoundsException
6. InputMismatchException

7. NegativeArraySizeException

8. ClassCastException

9. FileNotFoundException

10. ClassNotFoundException

9. Result

Thus, the Java program was successfully implemented and executed to demonstrate 10 different exception classes. The experiment helped in understanding exception handling using try-catch blocks, built-in exception classes, and the inheritance hierarchy of Java exceptions.